Year 13 Further Maths - Collisions

Similar questions - Walls, vectors and successive impacts

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A particle P of mass 2 kg moves on a large smooth horizontal plane and collides with a fixed smooth wall parallel to the unit vector $\mathbf{i}$. Before the collision its velocity is $(-5\mathbf{i}+12\mathbf{j})\ \mathrm{m\,s^{-1}}$ and after the collision it is $(-5\mathbf{i}-6\mathbf{j})\ \mathrm{m\,s^{-1}}$.

(a) Find the coefficient of restitution between P and the wall. **[2]**

(b) Find the impulse exerted by the wall on P, giving its magnitude and the angle it makes with $\mathbf{j}$. **[4]**

(c) Find the loss of kinetic energy. **[3]**

(d) Find the angle through which the direction of motion of P is turned. **[3]**

2 Two identical spheres A and B, each of mass m kg, move towards each other on a smooth horizontal surface along the same straight line, perpendicular to a fixed smooth vertical wall which is beyond B. Just before they collide, A and B have speeds $20\ \mathrm{m\,s^{-1}}$ and $10\ \mathrm{m\,s^{-1}}$ respectively. The coefficient of restitution between A and B is e.

(a) Show that the direction of motion of B is reversed in the collision. **[5]**

(b) B then collides with the wall; the coefficient of restitution between B and the wall is $\frac{1}{2}e$. There is no further collision between A and B. Determine the range of possible values of e. **[7]**

3 A particle P of mass 0.5 kg moves on a smooth horizontal floor with velocity $(8\mathbf{i}+6\mathbf{j})\ \mathrm{m\,s^{-1}}$. It strikes a smooth vertical wall along the y-axis (coefficient of restitution 0.5) and then immediately strikes a second smooth wall along the x-axis (coefficient of restitution 0.6), the walls forming a corner.

(a) Find the velocity of P after both impacts. **[4]**

(b) Find the total loss of kinetic energy. **[3]**

(c) Explain why the final velocity does not depend on the order of impacts. **[2]**

Question 1

(a) The wall is parallel to **i**, so the **j**-component is reversed and reduced: $e = \frac{6}{12} = 0.5$

(b) $\mathbf{I} = 2\left[(-5\mathbf{i} - 6\mathbf{j}) - (-5\mathbf{i} + 12\mathbf{j})\right] = -36\mathbf{j}$ N s: magnitude 36 N s, directed along $-\mathbf{j}$ (angle 180° with **j**), perpendicular to the wall

(c) KE before $= \frac{1}{2}(2)(25 + 144) = 169$ J; after $= \frac{1}{2}(2)(25 + 36) = 61$ J; loss $= 108$ J

(d) Before: $\tan^{-1}\frac{5}{12} = 22.6°$ from the **j** direction; after: velocity $(-5, -6)$ makes 39.8° with $-\mathbf{j}$. Direction turned through $180° - 22.6° - 39.8° = 117.6°$ (angle between $(-5, 12)$ and $(-5, -6)$: $\cos\theta = \dfrac{25 - 72}{13\sqrt{61}}$, $\theta = 117.6°$)

Question 2

(a) Taking A's direction positive: momentum $m(20) - m(10) = mv_A + mv_B$ so $v_A + v_B = 10$; restitution $v_B - v_A = e(30)$. Then $v_B = 5 + 15e > 0$ since $e \geqslant 0$ (and B was moving in the negative direction), so its direction is reversed

(b) $v_A = 5 - 15e$. After the wall B moves back with speed $\frac{e}{2}(5 + 15e)$, i.e. velocity $-\frac{e}{2}(5 + 15e)$ (towards A). No further collision if A is moving away from B at least as fast: $5 - 15e \leqslant -\frac{e}{2}(5 + 15e)$, i.e. $15e - 5 \geqslant \frac{e}{2}(5 + 15e)$, multiplying out: $7.5e^2 - 12.5e + 5 \leqslant 0$, $3e^2 - 5e + 2 \leqslant 0$, $(3e - 2)(e - 1) \leqslant 0$ so $\frac{2}{3} \leqslant e \leqslant 1$

Question 3

(a) First wall: $v_x \to -0.5 \times 8 = -4$, $v_y = 6$. Second wall: $v_y \to -0.6 \times 6 = -3.6$. Final velocity $(-4\mathbf{i} - 3.6\mathbf{j})$ m s^{-1}

(b) KE before $= \frac{1}{2}(0.5)(100) = 25$ J; after $= \frac{1}{2}(0.5)(16 + 12.96) = 7.24$ J; loss $= 17.76$ J

(c) Each wall only changes the component of velocity perpendicular to it: the x- and y-components are modified independently by factors -0.5 and -0.6, and multiplication commutes